

Weakly compact slice tensors form a minimal-tensor bimodule without injectivity

Candidate partial result for arXiv:1104.3812

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Status. Likely valid partial result. The theorem below removes the injectivity hypotheses from Corollary 2.3 of Runde’s source paper. It does not settle the product of two arbitrary weakly almost periodic elements.

1 Source question and metadata

The source is V. Runde, *Factorization of completely bounded maps through reflexive operator spaces with applications to weak almost periodicity*, J. Math. Anal. Appl. **385** (2012), 477–484, arXiv:1104.3812, DOI 10.1016/j.jmaa.2011.06.073 [1]. On PDF page 11 it asks:

Let (M, Γ) be a Hopf–von Neumann algebra. Is $\text{WAP}(M_*)$ a C^* -subalgebra of M ?

Here is the question as it appears in the source.

The following has been the primary motivation for the research in this paper:

Question. Let (M, Γ) be a Hopf–von Neumann algebra. Is $\text{WAP}(M_*)$ a C^* -subalgebra of M ?

Of course, one can replace $\text{WAP}(M_*)$ by $\mathcal{AP}(M_*)$ or $\mathcal{CAP}(M_*)$ in this question.

The only difficult part in answering the question is to determine whether $\text{WAP}(M_*)$ —or $\mathcal{AP}(M_*)$ or $\mathcal{CAP}(M_*)$ —is closed under multiplication. In [Run 2], the author showed that $\mathcal{CAP}(M_*)$ is indeed a C^* -algebra for injective M , and in [Daw 2], Daws proved for abelian M that both $\mathcal{AP}(M_*)$ and $\text{WAP}(M_*)$ are C^* -algebras.

We note:

Proposition 3.2. *Let (M, Γ) be a Hopf–von Neumann algebra. Then*

Figure 1: Source PDF page 11; the complete question and its immediate scope.

The source’s Corollary 2.3 states, under injectivity of M and N , that the weak-slice class $\mathcal{W}(M \bar{\otimes} N)$ is a bimodule over the spatial minimal tensor product. The result below shows that injectivity is not needed for this particular corollary.

2 Definitions

Let M, N be von Neumann algebras with their canonical preduals. For $u \in M \bar{\otimes} N$, define its left slice map

$$T_u : M_* \longrightarrow N, \quad T_u(f) = (f \otimes \text{id})(u).$$

Set

$$\mathcal{W}(M \bar{\otimes} N) = \{u \in M \bar{\otimes} N : T_u \text{ is weakly compact}\}.$$

The spatial minimal C^* -tensor product $M \otimes_{\min} N$ is viewed through its canonical isometric inclusion in $M \bar{\otimes} N$.

For $a \in M$, let $L_a(m) = am$ and $R_a(m) = ma$. These maps are normal, and their predual maps will be denoted

$$L_{a,*}(f) = f \circ L_a, \quad R_{a,*}(f) = f \circ R_a.$$

We use the same L_b, R_b notation for multiplication on N .

3 Proof intuition

Multiplication by an elementary tensor does not combine two uncontrolled weakly compact maps. It merely precomposes the old slice map with normal multiplication on M_* and postcomposes it with multiplication on N . Thus weak compactness survives for algebraic tensors. Norm density of the algebraic tensor product then gives the full minimal tensor product because weakly compact operators are norm closed.

For a Hopf coproduct, this produces a canonical multiplier algebra $A_\Gamma = \Gamma^{-1}(M \otimes_{\min} M)$. What remains open is precisely the case where both coproducts have weakly compact slice maps but neither lies in the minimal tensor product.

4 The partial theorem

Theorem 1 (injectivity-free minimal-tensor bimodule theorem). *For arbitrary von Neumann algebras M, N ,*

$$(M \otimes_{\min} N) \mathcal{W}(M \bar{\otimes} N) \subseteq \mathcal{W}(M \bar{\otimes} N) \quad \text{and} \quad \mathcal{W}(M \bar{\otimes} N) (M \otimes_{\min} N) \subseteq \mathcal{W}(M \bar{\otimes} N).$$

In particular, $\mathcal{W}(M \bar{\otimes} N)$ is a bimodule over $M \otimes_{\min} N$. No injectivity assumption on M or N is required.

Proof. First observe that $u \mapsto T_u$ is contractive from $M \bar{\otimes} N$ into $\mathcal{B}(M_*, N)$. Since weakly compact operators form a norm-closed operator ideal, $\mathcal{W}(M \bar{\otimes} N)$ is a norm-closed linear subspace.

Fix $u \in \mathcal{W}(M \bar{\otimes} N)$ and $a \in M, b \in N$. Directly on algebraic tensors, and hence on all of $M \bar{\otimes} N$ by normality, one has

$$T_{u(a \otimes b)} = R_b \circ T_u \circ R_{a,*}, \tag{1}$$

$$T_{(a \otimes b)u} = L_b \circ T_u \circ L_{a,*}. \tag{2}$$

Indeed, if $u = \sum_j m_j \otimes n_j$, then for $f \in M_*$,

$$T_{u(a \otimes b)}(f) = \sum_j f(m_j a) n_j b = R_b T_u(R_{a,*} f),$$

$$T_{(a \otimes b)u}(f) = \sum_j f(a m_j) b n_j = L_b T_u(L_{a,*} f).$$

All four multiplication maps in (1)–(2) are bounded. The operator-ideal property of weakly compact maps therefore implies $u(a \otimes b), (a \otimes b)u \in \mathcal{W}(M \bar{\otimes} N)$.

By linearity the same conclusion holds for every $z \in M \odot N$. Now take $z \in M \otimes_{\min} N$ and choose $z_k \in M \odot N$ with $\|z_k - z\| \rightarrow 0$. Multiplication in $M \bar{\otimes} N$ gives

$$\|uz_k - uz\| \leq \|u\|\|z_k - z\| \rightarrow 0, \quad \|z_k u - zu\| \rightarrow 0.$$

The norm-closedness noted in the first paragraph yields $uz, zu \in \mathcal{W}(M \bar{\otimes} N)$. $\square$

For completeness, the norm-closedness invoked above can be seen from the standard bidual criterion: if weakly compact $S_k : X \rightarrow Y$ converge to S in operator norm, then $S_k^{**} \rightarrow S^{**}$ in norm; since $S_k^{**}(X^{**}) \subseteq Y$ and the canonical copy of Y is norm closed in Y^{**} , one has $S^{**}(X^{**}) \subseteq Y$, so S is weakly compact.

Corollary 2 (Hopf–von Neumann consequence). *Let (M, Γ) be any Hopf–von Neumann algebra and put*

$$A_\Gamma = \{y \in M : \Gamma(y) \in M \otimes_{\min} M\}.$$

Then A_Γ is a unital C^ -subalgebra of M , and $\text{WAP}(M_*)$ is an A_Γ -bimodule.*

Proof. Since Γ is a unital $*$ -homomorphism and $M \otimes_{\min} M$ is a norm-closed C^* -subalgebra of $M \bar{\otimes} M$, its inverse image A_Γ is a unital C^* -subalgebra. By Proposition 3.2 of the source [1],

$$x \in \text{WAP}(M_*) \iff \Gamma(x) \in \mathcal{W}(M \bar{\otimes} M).$$

If $x \in \text{WAP}(M_*)$ and $y \in A_\Gamma$, then the theorem and multiplicativity of Γ give

$$\Gamma(xy) = \Gamma(x)\Gamma(y) \in \mathcal{W}(M \bar{\otimes} M), \quad \Gamma(yx) = \Gamma(y)\Gamma(x) \in \mathcal{W}(M \bar{\otimes} M).$$

Another application of the displayed equivalence proves $xy, yx \in \text{WAP}(M_*)$. $\square$

5 Verification notes

The proof uses only four standard facts: multiplication by a fixed von Neumann algebra element is normal; bounded pre- and post-composition preserve weak compactness; weakly compact operators are norm closed; and $M \odot N$ is norm dense in $M \otimes_{\min} N$. The two slice identities (1)–(2) have been checked in both orientations. No computational experiment is relevant or used.

An adversarial audit is included as `verification_report.md`. The recommended human check is to compare the slice conventions in (1)–(2) with source equation (1).

6 Limitations and novelty status

The original C^* -algebra question remains open: the theorem does not apply to a product of two arbitrary WAP elements unless one coproduct lies in the minimal tensor product. For noninjective M no identification of A_Γ with $\text{CAP}(M_*)$ is asserted.

Cheap run indexes and bounded web searches on 19 July 2026 found no exact version of the injectivity-free theorem. Runde states Corollary 2.3 with injectivity. Daws’s later treatment [2] still describes the Hopf WAP algebra question as unresolved; its counterexample concerns the unconstrained two-fold weak-slice class and does not supply a Hopf coproduct counterexample. The result is therefore marked *likely valid partial result*, not literature-certified novel.

Human-review recommendation. Promote as a substantial hypothesis-removal partial theorem after an expert independently verifies the two slice identities and the canonical minimal tensor inclusion.

References

- [1] V. Runde, *Factorization of completely bounded maps through reflexive operator spaces with applications to weak almost periodicity*, J. Math. Anal. Appl. **385** (2012), 477–484; arXiv:1104.3812; DOI: 10.1016/j.jmaa.2011.06.073.
- [2] M. Daws, *Non-commutative separate continuity and weakly almost periodicity for Hopf von Neumann algebras*, arXiv:1409.7302 (2014; revised 2015).